Gmark IoT

1. Problem Statement

Commercial facilities, residential societies, and industrial plants face severe operational bottlenecks due to unmonitored equipment, manual meter readings, delayed fault detection, and fragmented asset records. Critical systems such as hydro-pneumatic (HYPN) pressure boosters, sump drainage pumps, sewage treatment plants, and emergency fire sprinkler panels frequently encounter unnoticed pressure drops, cavitation, dry runs, electrical overload, and pump burnouts. Without centralized cloud supervisory monitoring and structured 4M (Man, Machine, Material, Method) governance, organizations suffer from high energy and water wastage, unplanned downtime, costly asset damage, and lack of auditable maintenance records.

2. Our Solution

Gmark IoT provides a comprehensive Industrial IoT (IIoT) SCADA automation and 4M asset management platform. It seamlessly connects field hardware controllers (PLCs, AIPCU controllers, Modbus devices) with modern cloud supervisory software. Gmark IoT delivers real-time machine telemetry (delivery/suction line pressure, flow rates, frequency, runtime hours, operational status), animated process flow diagrams, automated threshold alerts with audio sirens, preventive maintenance scheduling, automated equipment code sequencing, and granular role-based access control.

3. Project Overview

Gmark IoT is an all-in-one industrial automation and asset governance platform tailored for commercial facilities, industrial plants, and residential societies. It connects physical hardware controllers with digital supervisory control. The system features an interactive SCADA dashboard with live animated pipe networks and pump units, a centralized multi-tenant facility hierarchy (Main Facility, Sub-Facility, Tower Building), automated equipment code sequencing, and platform-wide configuration management.

4. Goals

Facility & Plant Managers

- Provide real-time remote visibility and supervisory control over critical pumps, motors, and fire panels.
- Prevent catastrophic equipment failures and utility supply interruptions through predictive threshold alarms.
- Increase operational efficiency, uptime, and energy savings with automated runtime telemetry.
- Establish a centralized, cloud-accessible digital twin of multi-facility operations.

Maintenance Teams & Engineers

- Streamline equipment maintenance workflows, spare parts management, and AMC service agreements.
- Standardize asset tracking with automated, non-duplicating Machine and Asset sequence coding.
- Receive instant audible sirens and visual badge notifications for pump trips, overload, and dry-run faults.
- Eliminate error-prone manual physical logbooks with automated time-series telemetry recording.

Societies & Commercial Enterprises

- Guarantee 24/7 unhindered domestic water supply and emergency fire-readiness across residential towers.
- Accurately allocate operational expenses, power bills, and maintenance costs to designated Cost Centers.
- Enforce strict operational security through Role-Based Access Control, ensuring authorized personnel access only assigned devices.
- Extend equipment lifespan and reduce capital expenditure by proactively maintaining optimal operating parameters.

5. Vision

To pioneer intelligent, zero-downtime industrial and facility automation by transforming physical machines into transparent, connected digital assets—ensuring safety, sustainability, and operational peace of mind for communities and enterprises worldwide.

6. Our Values

- **Empowerment:** Enable facility managers, engineers, and operators to make data-driven maintenance decisions through modern digital tools.
- **Reliability:** Ensure uncompromised 24/7 continuous operation and fail-safe monitoring for mission-critical water and fire networks.
- **Transparency:** Build operational trust through instantaneous, verified sensor telemetry and tamper-proof event logs.
- **Sustainability:** Promote environmentally responsible utility management by reducing energy wastage and conserving water.
- **Technology for Good:** Make complex industrial IoT technology simple, visual, and accessible for everyday operators and staff.
- **Community Safety:** Guarantee round-the-clock emergency fire readiness, pressure stabilization, and automated trip protection.
- **Scalable Innovation:** Build open, modular IoT architectures capable of effortlessly scaling from single pump sets to nationwide industrial hubs.

7. What the System Does

Area	*Purpose*
Live SCADA Monitoring	Real-time visual monitoring of pump systems, suction/delivery pressures, line frequencies, and run hours with animated flow indicators.
4M Machine Master	Register, track, and manage machine profiles, auto-generated sequence codes (MCH, AST), specifications, and parent-child location hierarchies.
Facility Hierarchy	Multi-level geographical organization connecting Main Facilities, Sub-Facilities, Societies, and Tower Buildings.
Threshold & Alert Engine	Monitors configurable high/low setpoints on PLC registers and triggers real-time audio alarms, visual warning badges, and acknowledgement logs.
Interactive Process Flow	Dynamic, animated SVG schematic diagrams representing physical booster stations, valves, pipes, and pump units in real time.
User & Role Governance	Strict Role-Based Access Control (RBAC) with customizable permissions across Super Admin, Society Admin, Building Admin, Operator, and Viewer.
Employee Management	Centralized staff master tracking employee codes (EMP), designations, contact records, and machine ownership assignments.
Historical Data & Analytics	Time-series logging of device register metrics, pressure trends, runtime histories, and exportable operational audit logs.
IoT Hardware Connectivity	High-throughput telemetry ingestion supporting MQTT message brokers and industrial Modbus TCP/RTU protocol controllers.

8. Platform Ecosystem

- **Central Cloud SCADA Web Portal –** an interactive web application for facility managers and administrators to view live plant schematics, monitor telemetry, and supervise multi-site infrastructure.
- **4M Industrial Asset Suite –** a centralized administrative module managing machines, locations, maintenance heads, cost centers, specifications, and warranty contracts.
- **Edge IoT Telemetry Broker –** a robust message broker handling bi-directional MQTT data packets and register sync between field controllers and the cloud.
- **Role-Based Operations Console –** tailored user views empowering society administrators, field engineers, and building operators with dedicated action menus.

9. Key Features

- Live Animated SCADA Process Flow Diagrams
- Real-time Multi-line Pressure & Flow Monitoring
- Dynamic Pump Status Indicators (Auto / Manual / Run / Trip / Standby)
- Multi-tier Threshold Alarms with Real-time Audio Alerts
- Automated Machine & Asset Code Sequencing (MCH / AST)
- Multi-level Geographical Facility & Building Hierarchy
- 4M Machine Master with Technical Specification & Nameplate Capture
- Cost Center & Departmental Financial Budget Allocation
- Employee Master with Role Assignments & Ownership Linking
- Granular Role-Based Access Control (RBAC)
- Time-Series Sensor Analytics & Historical Logging
- Industrial MQTT & Modbus Controller Protocol Integration